Anomaly Detection

So far, you have learned many concepts that are there in Machine Learning

The purpose of this lecture is to let you know how you can take the concepts that you already know like Random Forests, SVMs, KNN, etc., and modify them for using them for another purpose rather than classification.

What is an Anomaly?

- Anomaly is synonymous with an outlier. These terms are often interchanged and may be called Novelty depending on the context.

What's the difference?

- Anomaly means something which is not a part of the normal behavior
- Novelty means something unique, or something that you haven't seen it before(novel)

Applications of Anomaly/Outlier/Novelty Detection:

- Credit Card Fraud Detection
- System Intrusion Detection
- Ecosystem Disturbances in Weather and Environment (alarming about Tsunami, Earthquake, etc.)
- You can read how some startups are thinking out of the box using these techniques from a blog here.

1. Distribution Based

- The simplest way for detecting an outlier would be to use distribution parameters (mean and standard deviation).
- Consider a feature X with some observations $x_1, x_2, \ldots, x_n$ with some outliers.
- We know that it will follow some distribution which will have parameters θ.
- Let follow a Gaussian distribution with some mean(μ) and standard deviation(σ)
- If we know the distribution of the data, we'll try to fit the distribution. However, the problem arises with the distribution parameters.
- While we know the distribution, the parameter estimates of the distribution are often corrupted by the noise/outlier
- Hence, we need to robustly estimate the parameters of the distribution. We do this using an algorithm called RANSAC which stands for Random Sample Consencus

2. Random Sample Consensus (RANSAC)

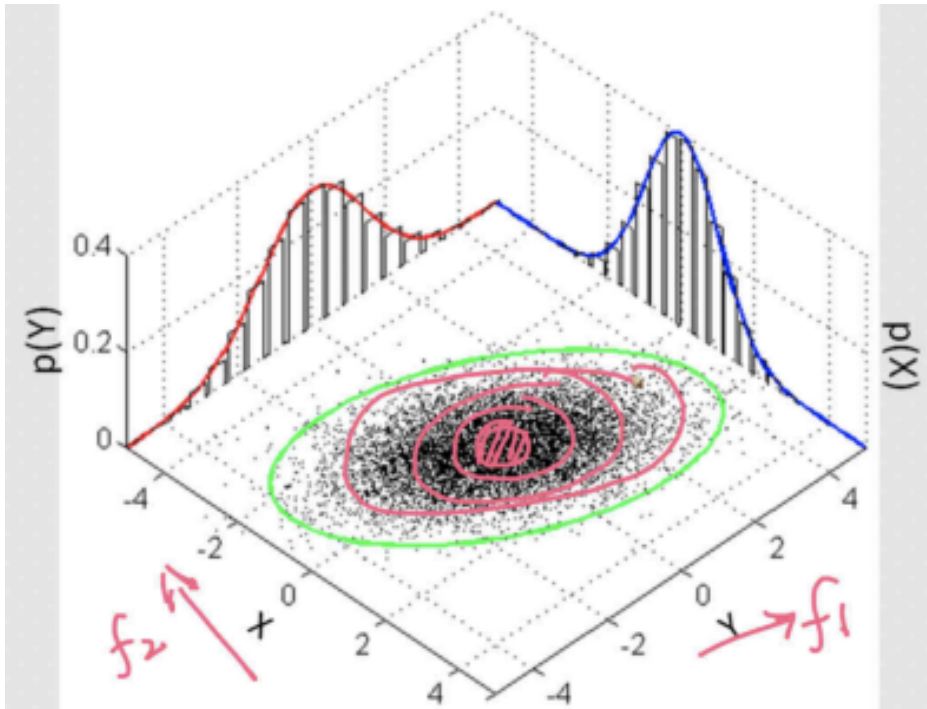
- RANSAC is a trial-and-error approach that works very well in real life.
- Imagine a dataset X with a number of points having parameters μ and σ^2. Let's call them collectively θ
- There are mainly three steps involved in RANSAC. They are
    - Sample a subset of points from the dataset (n'). We consider this point as an inlier.
    - Now, compute a model that estimates the parameters of the sampled points.
    - Score the model which indicates how many points will support the model.
- We repeat these three steps iteratively and then select the model best supported by the data, which then tells which points are inliers and which are outliers.

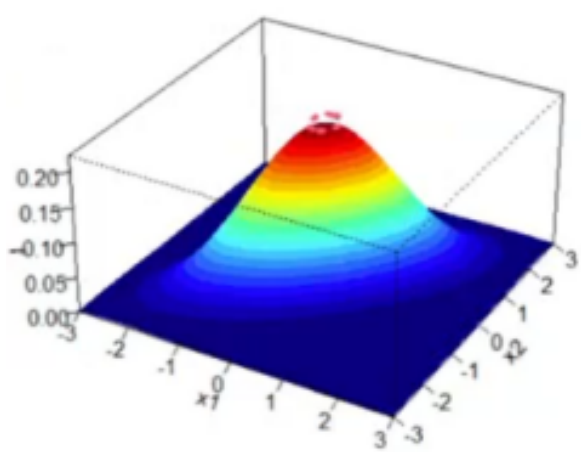
Extending the idea to higher dimensions

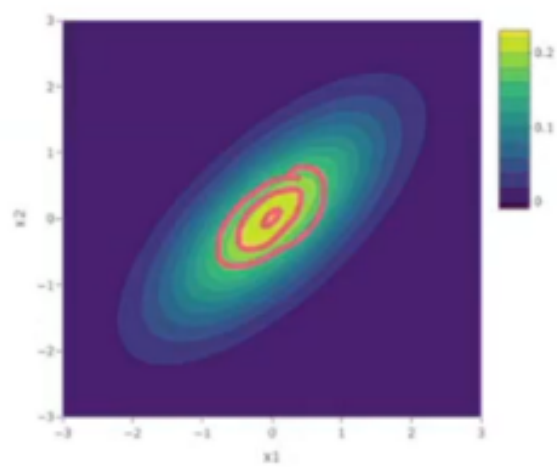
- Till now, we assumed that we have only a single feature X which follows Gaussian Distribution.
- Now, imagine we have d-dimensional data where each point $x_i \in R^d$ and the data is not labeled.
- If we know the data points x_is follow multivariate Gaussian distribution(unimodal), then X follows normal distribution; $X \sim (\vec{u}, \Sigma)$, where is $\vec{u}$ mean vector and Σ is a covariance matrix
- Here we'll consider $(\vec{u}, \Sigma)$ as θ
- If you remember from GMMS, in multidimensional space, the shape of gaussian was similar to a hill where the density of the points was highest in the middle contour, and it keeps getting low as we move away from the center
- In this case too, RANSAC can be applied. Farther away from centroid, we'll know that it is an outlier.
- Similar principle is used in another method called Elliptical Envelope.

3. Elliptic Envelope

- We know that a Unimodal Multivariate Gaussian Distribution on a single plane will look like ellipses if visualized on a plane. This idea can be extended to find out an outlier







- Given some data X where x_is Є R^d and X follows Normal Distribution being unimodal, Elliptical Envelope robustly estimate the parameters ($\vec{u}$, Σ).
- The term robustly means without getting impacted by outliers
- Next, then we remove the points that are outliers which are very far away from the centroid

What if the distribution is non-gaussian? Do we need to convert it into Gaussian distribution?

- While the elliptical envelope method assumes that the distribution is Gaussian, the strategy can be applied to any distribution.

- As long as we know any distribution and its parameters θ, we can extend our strategy to use RANSAC and estimate parameters θ.

- The can be other distributions such as multivariate Poissons, multivariate log normals, etc., but we don't use them as much.

- One other strategy is to convert them into Gaussians but we don't necessarily have to.

- As long as there is any distribution we can calculate the probability of x_i Є X using PMF/PDF.

Disadvantages:

- It cannot be used for non-unimodal data

- It is specifically for multivariate Gaussians

- If the data fails to meet the assumptions of unimodal and multivariate gaussian, the whole thing crashes.

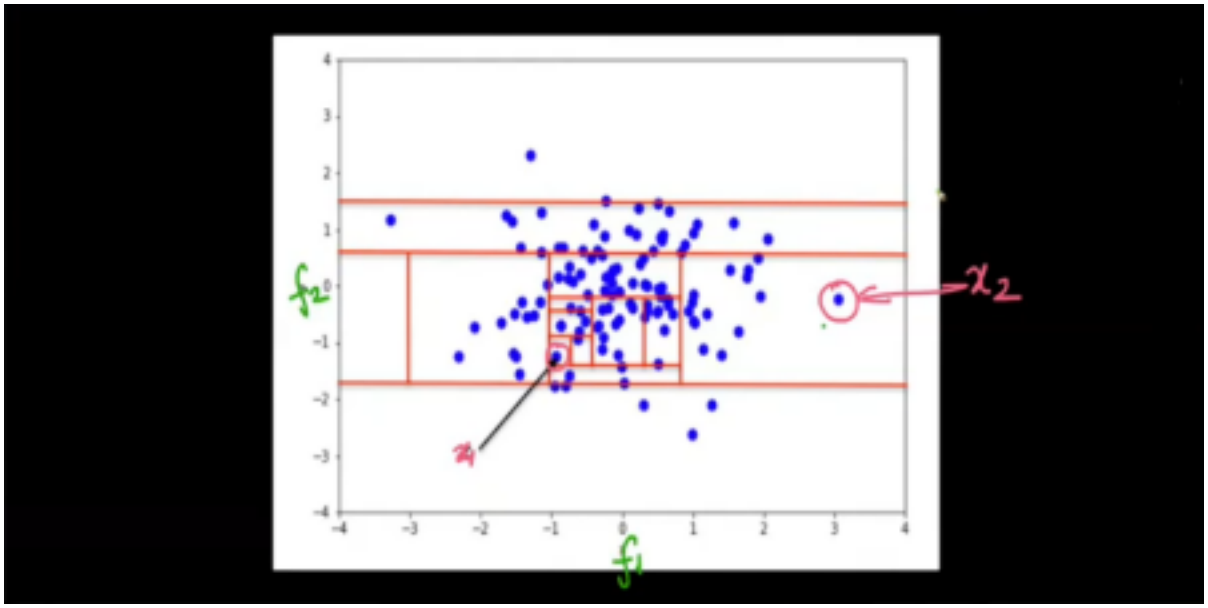
4. Isolation Forests

- Consider a dataset D which contains data points x_1, x_2,...., x_n. Just like Random forests, Isolation Forests build many trees.

- These are the steps involved in Isolation Forest:
    - Build many trees like random forests
    - For each tree:
        - Randomly pick a feature
        - Randomly threshold that features
        - Build each tree until the leaf consists of only one datapoint

- Isolation Forests are also known as iForests

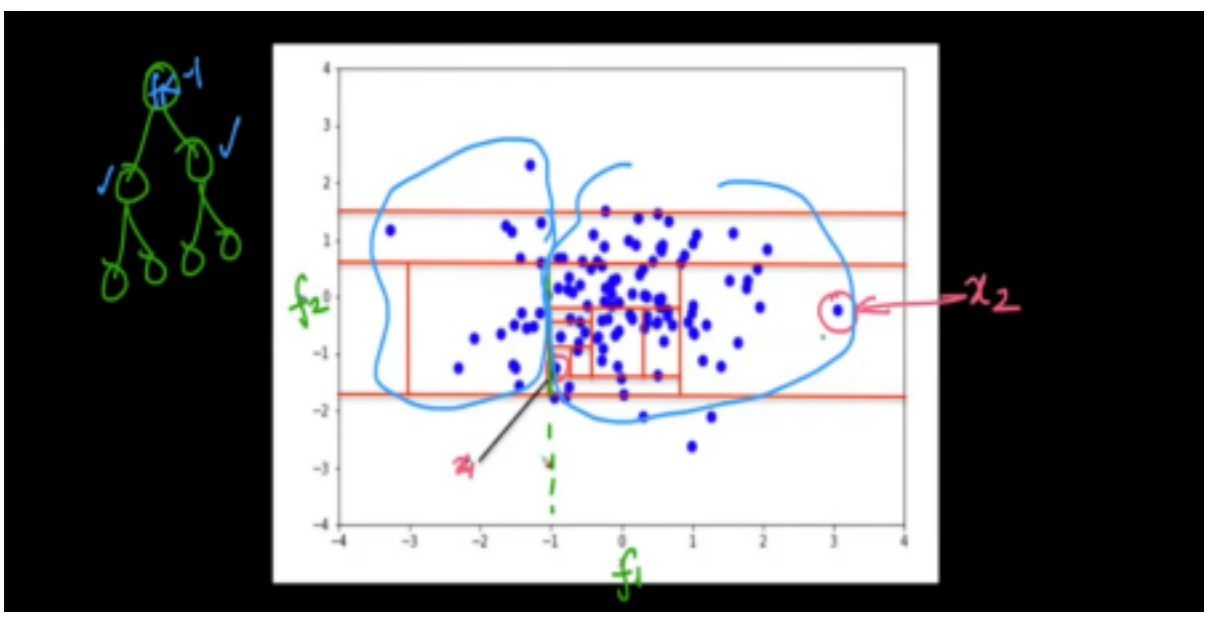
- Consider the plot along feature f_1 and f_2 given below:



- In isolation forests, we are building totally random trees. So if we pick feature f_1 and put a threshold there will be a vertical bar.

- Similarly, if we pick feature f_2 and put a threshold there will be a horizontal bar.

- For example, if we pick feature f_1 and we select threshold as $f_1 < 1$, then our first root node will be based on this condition



- Based on the diagram above,
    - The node containing x_1 will be at more depth.

    - Observe that the point x_1 is in dense region, and point x_2 is far away

    - That is because, to break the point x_1 from all the other points, more and more splits will be required and that will increase the depth of the node containing point x_1.

- So, to sum it up, the idea behind Isolation forest is:
    - On an average outliers have lower depth in the random trees
    - On an average, inliers have more depth in the random trees

Evaluation of Isolation Forest

- Imagine, we have build random trees. For each point x_i in the dataset, we can get an average depth.

- We use this average depth to convert into a metric.

- Apart from this, there are lot of different metrics, that people have came up with over the years

- But, the basic intuition is that lesser the average depth, higher likelihood is there that it is an outlier

Deciding average depth of a point

- There is no one metric specifically used for average depths in iForests. At the end, whichever metric you use, it is based on the threshold.

- There are a lot of metrics that researchers have came up with over the years.

- But, studying them in this lecture is out of scope.

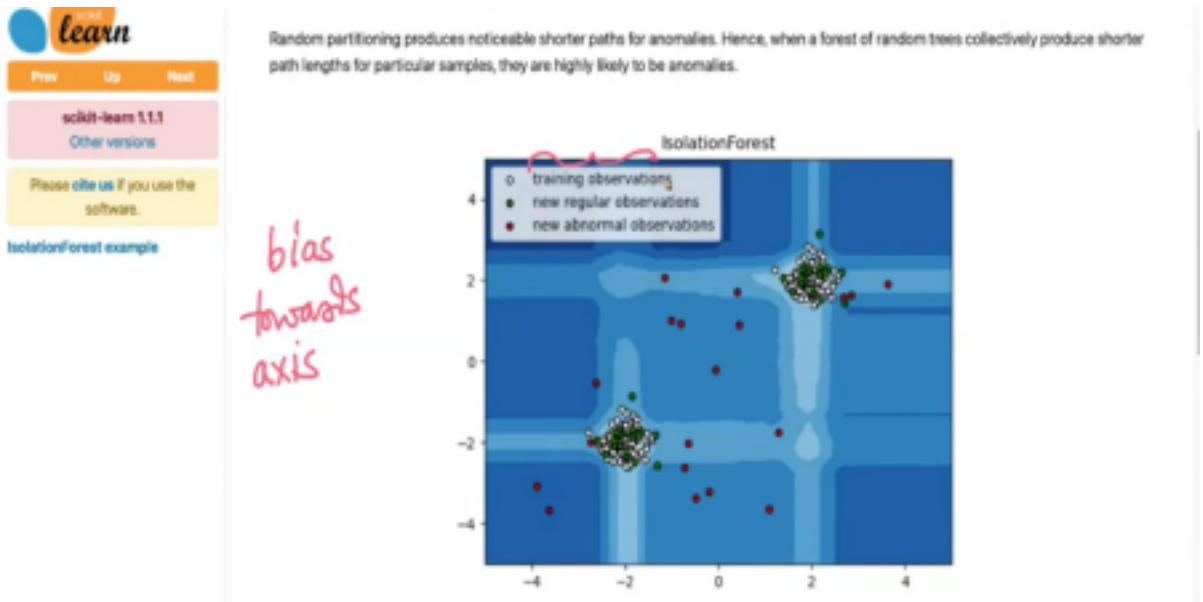
Disadvantages

- One of the major limitation of iForests is that they are biased towards axis parallel splits.

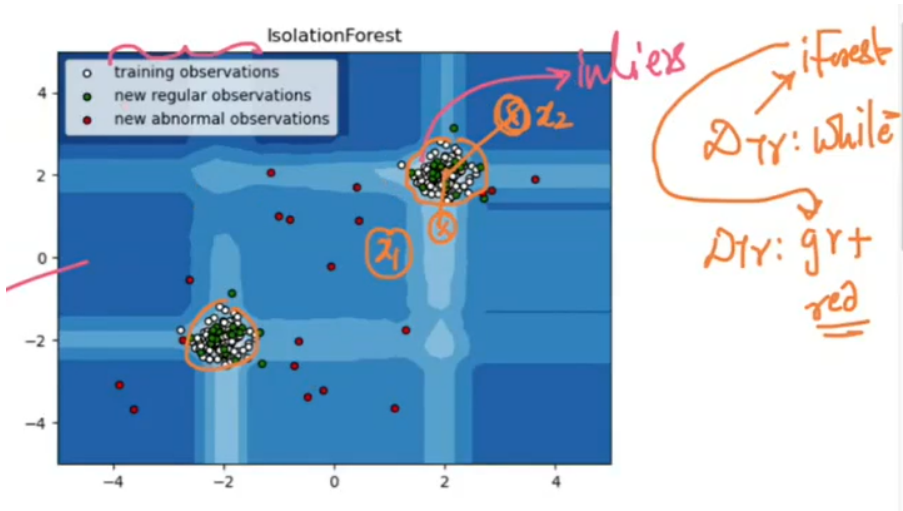
- iForests makes splits and these splits are always parallel to either of the axis.

- Because of this, the boundary will not be smoothed.
    - In the diagram given below, the different shades of blue represents the likelihood of a point to be an outlier. Darker the color, it is more likely that the point in that region will be an outlier

- We've trained the iForest model using training data(white points)
- It is tested on testing data(red + green) where red color indicates outliers



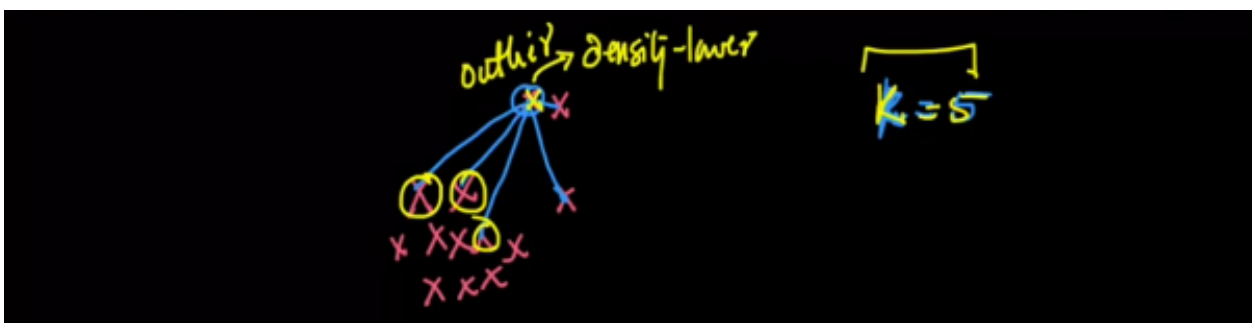
- Now imagine two points x_1 and x_2 as shown in the diagram given below.
- Both the points are almost equidistant from the nearest cluster. x_1 is on the axis and point x_2 is off axis.
- Because the model is biased towards axis parallel splits, it will classify point x_1 as an inlier and x_2 as an outlier
- This is also known as banding in signal processing



5. Local Outlier Factor (LOF)

- On a higher level, LOF is based on two ideas: KNN and density
- The core idea behind LOF is to compare the density of a point with its neighbors' density
- If the density of a point is less than the density of its neighbors, we flag that point as an outlier

- Imagine a bunch of datapoints as shown below



- We compute the density of a point based on average distance.
- If average distance between a point and its **K** nearest neighbors is large, it is more likely that the point will be an outlier
- Also, larger the value of **K**, more confident are the results.

Some concepts to understand the working of LOFs:

1(a) K-distance

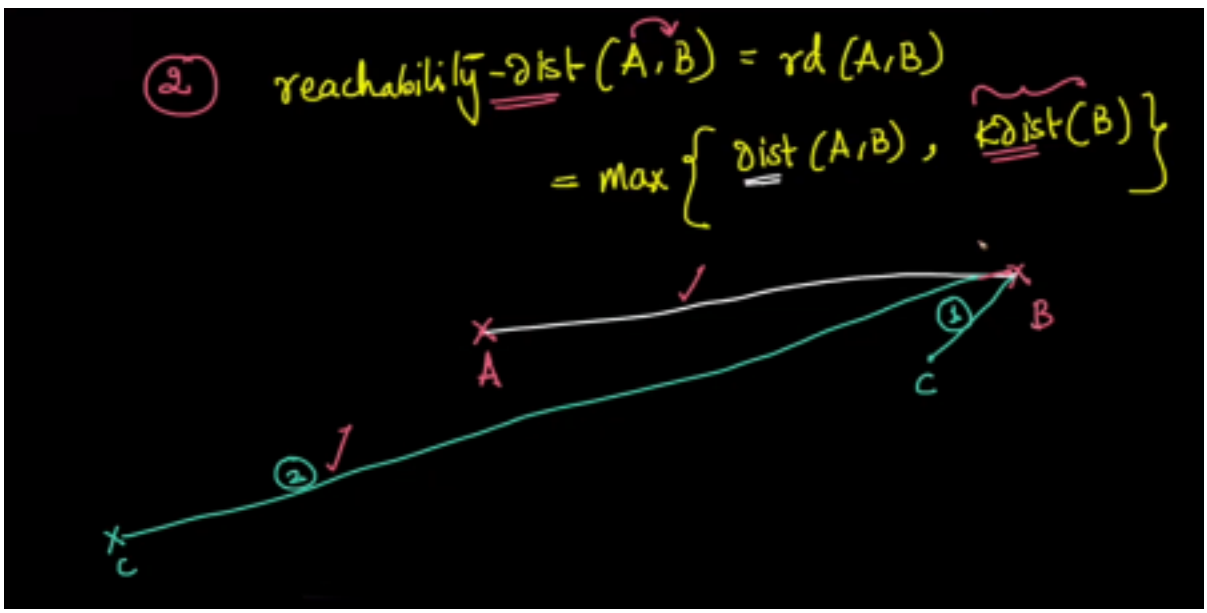
- We define K-distance of a point **A** as the distance of point **A** to its **K**th nearest neighbor
- In general, larger the value of k-distance is, farther away the points is from other datapoints

1(b) Set: Nk (A)

- It is a set of k-nearest neighbors of point **A**.

2. Reachability distance

- From point **A** to point **B** , we define reachability distance as maximum of the distance from point **A** to point **B** and the maximum k-distance of point **B**
- Consider point **B** with some **k** nearest neighbors shown in the diagram below.



- There is a possibility that some neighbors might be close(condition 1) and some neighbors might be very far away(condition 2)
- In this case, there is a neighbor of point **B** whose k-distance is greater than the distance between point **A** and **B**, and hence, it is considered as its reachability distance.

3. Local Reachability Density

- It is often represented as **$lrd_k(A)$** , which tells local reachability density of a point **A**
- It is defined as the average reachability distance between point **A** and **k** neighbors
- So, $lrd_k(A) = \frac{\sum_{B \in N_k(A)} rd_k(A,B)\$}{N_k(A)}$
- The summation in above equation represents the sum of reachability distances from a point **A** and set of neighbors **B** as **B** Є **$N_k(A)$**

- We define Local Outlier Factor of point as follows:

$$LOF_k(A) = \frac{\sum_{B \in N_k(A)} lrd_k(B)}{|N_k(A)|.ldr_k(A)}$$

  - **$ldrk_k(A)$** is the density of point **A**
  - The expression $\frac{\sum_{B \in N_k(A)} lrd_k(B)}{|N_k(A)|}$ is the average neighborhood density
  - So, LOF of point **A** is nothing but the average neighborhood density(lrd) of point **A** divided by the density of **A**

Interpretation of LOF

- If **LOF(A) = 1**, then we can say that the point has same density(lrd) as its **k** nearest neighbors

- If **LOF(A) > 1**,, then the **k** neighbors of point **A** have higher density than point **A**.
  - That does not mean point **A** is an outlier. It may or may not be.
  - But if **LOF(A) >>> 1**, then the point is definitely an outlier.

- If **LOF(A) < 1**, then the point has more density than its nearest neighbors.

Disadvantages of LOF

- Finding optimal K

- Finding threshold.
  - If LOF(A) >> 1, what is the threshold??